

# OTC Integration v2

Juyoung Park, 2026.

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Q1 A bowl of hot food is placed on a table to cool. The room temperature is constant at  $20^{\circ}\text{C}$ . The rate at which the temperature of the food cools is proportional to the difference between the temperature of the bowl and the temperature of the room. The temperature of the bowl when placed on the table was  $80^{\circ}\text{C}$ . After two minutes the temperature of the bowl was  $70^{\circ}\text{C}$ . The food will be cool enough to eat when the temperature has fallen to  $56^{\circ}\text{C}$ . How long will it be from the time the bowl is placed on the table until the food is sufficiently cool to eat?

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Q2 The model for the height  $h$  metres, of a tree  $t$  months after being planted, says that while the tree is growing, the rate of increase of height with respect to time is  $k\sqrt[3]{9-h}$  metres per month. When  $t = 0, h = 1$ , and the rate at which the tree is growing is  $0.4\text{m}$  per year. Find the time it takes for the tree to achieve its maximum height.

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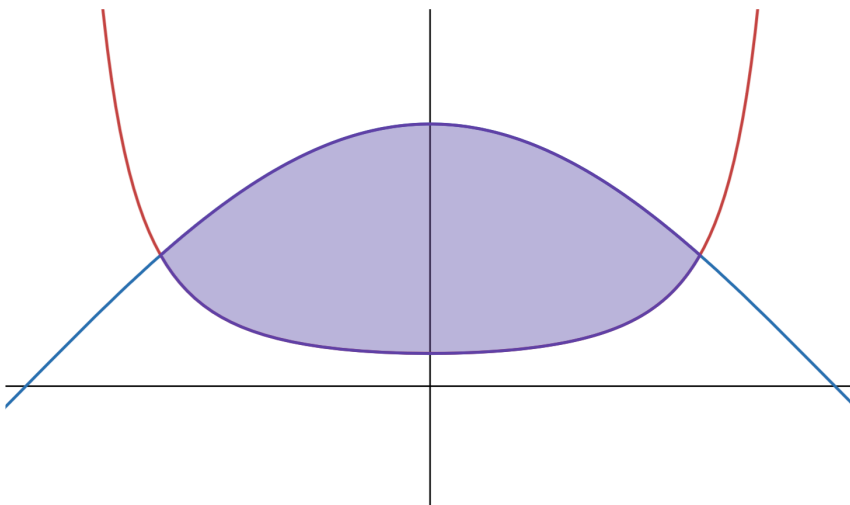
Q3 Find the value of  $x$  when  $y = 9$  if

$$\frac{dy}{dx} = \frac{8ye^{2x}}{1 + e^{2x}}$$

and  $y = 4$  when  $x = 0$ .

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Q4 The diagram below shows the graphs of  $y = 8\cos(kx)$  and  $y = \sec^2(kx)$  for the same constant  $k$ . Find the shaded area bounded by the curves.



Q5 The rate at which a body loses velocity at any instant as it passes through a medium is given by  $-kv\frac{m}{s^2}$ , where  $v$  is the velocity of the body at that instant and  $k$  is a positive constant. If its initial velocity is  $u\frac{m}{s}$ ,

show that the time taken for the body to halve its velocity is

$$\frac{\ln 2}{k}$$

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Q6 A dog's acceleration in  $\frac{m}{s^2}$  is given by  $a(t) = (kt + 1)^{-2}$ , where  $m$  is the distance in metres from a stick and  $t$  is the time in seconds. Initially, the dog's velocity is  $4\frac{m}{s}$  and it is 2 metres from the stick. Find an expression for the dog's distance from the stick after  $t$  seconds.

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Q7 Find an explicit expression for  $y$  in terms of  $x$ , if

$$(x + 2)\frac{dy}{dx} = (2x^2 + 4x + 1)(y - 3)$$

and  $y = 7$  when  $x = 0$ .

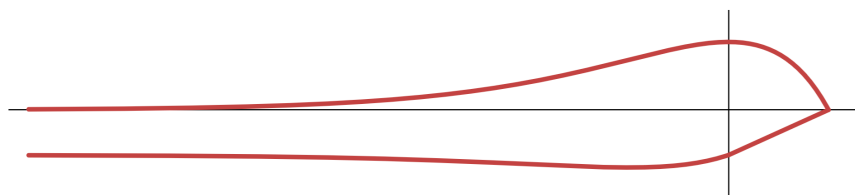
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Q8 Engineers are designing a new style of aircraft wing. The shaded area below shows the latest design from a cross section view. It comprises of the following functions

$$y_1 = -0.3(x - 3)e^x, \{-7 < x < 1\}$$

$$y_2 = 0.15e^x - 0.2, \{-7 < x < 0\}$$

$$y_3 = 0.2x - 0.2, \{0 < x < 1\}$$



Find the cross sectional area of the new wing design.

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Q9 Find the general solution of the differential equation

$$4(x^2 + 3)\frac{dy}{dx} = x(16 - y^2)$$

If  $y = 2$  when  $x = 1$ , find the value of the constant of integration.

HINT:

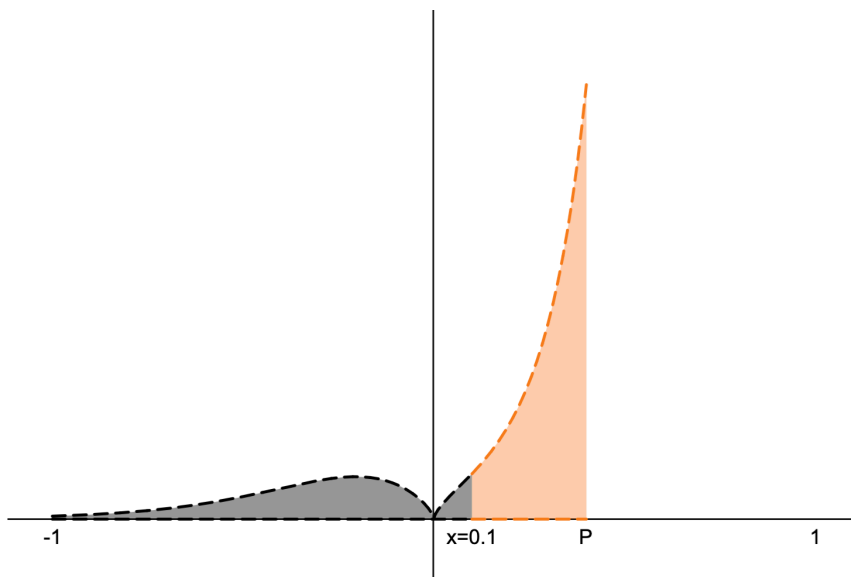
$$\frac{a}{a^2 - x^2} = \frac{P}{a + x} + \frac{Q}{a - x} \implies P = Q = \frac{1}{2}$$

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Q10 The following graph shows the function

$$y = x^{\frac{2}{3}} \left(1 - x^{\frac{5}{3}}\right)^{-6}$$

from  $-1 < x < p$ . The two shaded areas from  $-1 < x < 0.1$  and  $0.1 < x < p$  are equal in size. Calculate the value of  $p$ .



Q11 Find  $k$  such that the area enclosed by the lines  $x = 0$ ,  $x = 5$ , the  $x$ -axis, and the curve

$$y = e^{kx}$$

is greater than

$$\frac{5}{k}$$

Q12 Find the value of  $x$  when  $y = 9$  if

$$\frac{dy}{dx} = \frac{6ye^{3x}}{1 + e^{3x}}$$

and  $y = 4$  when  $x = 0$ .

Q13 The number of germs in a petri dish  $t$  hours after the start of observations is denoted by  $N$ . The variation in the number of germs is modelled by the differential equation

$$\frac{dN}{dt} = kN$$

where  $k$  is a constant and  $N$  is taken to be a continuous variable measured in millions. It is given that  $N = 50$  when  $t = 0$ . Given also that  $N = 100$  when  $t = 10$ , find how many hours pass before the number of germs in the petri dish has  $N$  exceeding 1000.

Q14 A patient has a glucose solution administered intravenously into his bloodstream at a constant rate,  $r$ . As the glucose is added, it is converted into other substances and removed from the bloodstream at a rate that is proportional to the concentration at the time.

A model  $C(t)$ , the concentration of the glucose solution in the bloodstream, is given by

$$\frac{dC}{dt} = r - kC$$

where  $k$  is a positive constant.

At time  $t = 0$ , the concentration is  $C_0$  where  $C_0 < \frac{r}{k}$ .

Find an expression for the concentration  $C(t)$  at any time, and explain what happens as  $t$  increases.

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Q15 Find the area between the curve

$$y = \cos^2 x$$

and the  $x$ -axis,  $n\pi \leq x \leq (n\pi + 1.5\pi)$ .

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Q16  $A$  is the area enclosed by the graph of the function  $y = f(x)$ , the lines  $x = a$ , and  $x = b$ , and the line  $y = c$  where  $c > f(x)$  for all  $x$  satisfying  $a < x < b$ .

Using all integrals involved, prove that

$$A = c(b - a) - \int_a^b f(x) dx$$

It may be assumed that  $f(x) > 0$  for all  $x$  satisfying  $a < x < b$ .

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Q17 The number of radioactive particles in a substance  $t$  years after the start of observations is given by  $N$ . The variation in the number of radioactive particles is modelled by differential equation

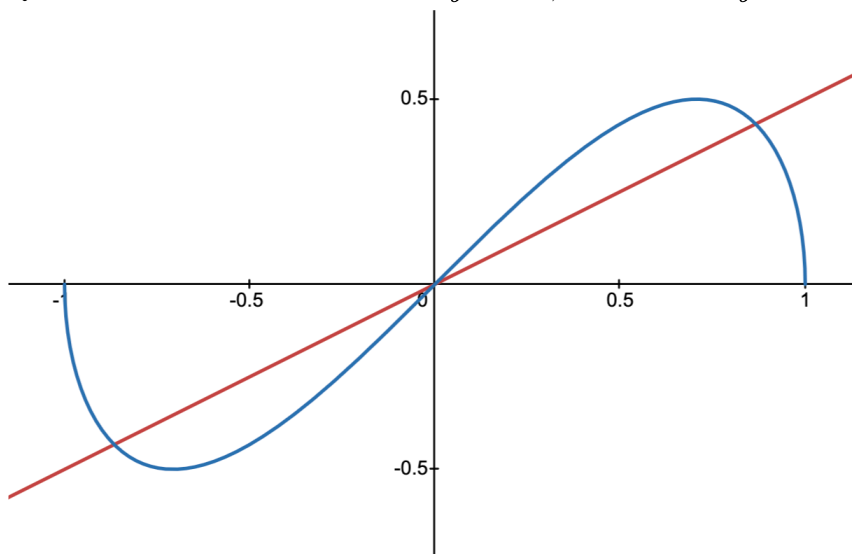
$$\frac{dN}{dt} = kN$$

where  $k$  is a constant, and  $N$  is taken to be a continuous variable measured in billions. It is given that  $N = 25$  when  $t = 0$ .

i) Solve the differential equation, obtaining a relation between  $N$ ,  $k$ , and  $t$ . ii) Given also that  $N = 10$  when  $t = 5$ , find the value of  $k$ . Hence find the time that it takes for the number of radioactive particles to halve.

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Q18 Find the areas between the line  $y = 0.5x$ , and the curve  $y = x\sqrt{1-x^2}$  as shown below.



Q19 The height  $h$  in  $cm$ , of the fluid in a container at time  $t$  seconds, is given by the differential equation

$$\frac{dh}{dt} = \frac{A}{-\sqrt{h^3}}$$

where  $A$  is a constant. Initially, the height of the fluid is  $H$   $cm$ , and after 60 seconds, the container is empty. Find whether the time required for the height of fluid to be  $\frac{H}{2}$   $cm$  is more or less than 30 seconds, and by how much more or less.

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